

Two Blind Observables on a Signed Gauge Decomposition: An Invariance-Audited Diagnostic Reading of the Turok–Boyle Primordial Spectrum

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Abstract

Turok and Boyle compute the amplitude and spectral tilt of the primordial scalar perturbations from the high-temperature trace-anomaly coefficient of the Standard Model, $c_\beta = \frac{125}{108}\alpha_Y^2 - \frac{95}{72}\alpha_2^2 - \frac{49}{6}\alpha_3^2$ [Turok and Boyle, 2023], a coefficient they in turn quote from Arnold and Zhai [1995], evaluated at Planck-scale couplings taken from Buttazzo et al. [2013]. This note records a diagnostic reading of that object, and states plainly what is classical mathematics, what is imported physics, and what is newly applied here. The physical question is: what do the two observables Turok and Boyle extract actually reveal about the six-term signed sum that produces c_β , and what do they necessarily conceal? The amplitude, proportional to c_β^2 , reads only the magnitude of the net signed sum across gauge channels and is blind both to that sum’s internal decomposition and to its sign. Under the heuristic assumptions Turok and Boyle use to connect spatial wavelength to renormalization-group running, the tilt reads the dominant $SU(3)$ running combination $b_3\alpha_3$ alone. Neither observable resolves the channel decomposition. We quantify what is unresolved using the classical signed- L^1 descriptors $c_{\text{net}} = \sum_i c_i$ (total signed mass), $c_{\text{gross}} = \sum_i |c_i|$ (total variation mass), and the cancellation index $\kappa_{\text{cancel}} = 1 - |c_{\text{net}}|/c_{\text{gross}}$. The mathematics here is not new: reading c as a discrete signed measure, c_{net} and c_{gross} are exactly the Jordan-decomposition total signed and total variation masses [Bogachev, 2007], and $c_{\text{gross}}/|c_{\text{net}}|$ is exactly the classical *condition number of a sum* that governs floating-point summation error [Higham, 1993], so $\kappa_{\text{cancel}} = 1 - 1/\text{cond}$ is a bounded, invertible reparameterization of that textbook quantity, not a new invariant. What is applied here, apparently for the first time to this coefficient, is a systematic identifiability audit built on that classical vocabulary: we prove that only the net is fixed once the theory, truncation, and scale are fixed, while c_{gross} and κ_{cancel} are functions of the sign-partition alone, satisfying $\kappa_{\text{cancel}} = 2 \min(P, N)/(P+N)$ for the positive and negative masses P, N , monotonically non-decreasing under any refinement crossing the sign boundary, and admitting no decomposition-free value. Applied to Turok and Boyle’s own coefficients, the gauge-factor decomposition gives $\kappa_{\text{cancel}} \approx 0.20$; refining each channel into its gauge-boson and matter parts — a split Turok and Boyle display for $SU(3)$ and which we show holds uniformly — leaves the net unchanged but raises κ_{cancel} to 0.64; evaluating the same expression at the electroweak scale gives $\kappa_{\text{cancel}} \approx 0.002$. So $\kappa_{\text{cancel}} \approx 0.20$ is a descriptor of one stated decomposition at one stated scale and truncation, not a property of the Standard Model. We also distinguish, throughout, between algebraic constructions used only to witness non-identifiability (an arbitrary two-component split, a global sign flip) and the one physically motivated refinement available here (gauge boson versus matter, at fixed order and scale); the former are not alternative physically realizable decompositions of c_β . No claim is made about the validity of the Turok–Boyle construction, which remains contested in the refereed literature [Cline and Hell, 2026].

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1 Purpose, scope, and non-claims

Turok and Boyle propose that quantum zero-point fluctuations of dimension-zero fields seed the primordial scalar perturbations, and compute both the amplitude and the spectral tilt in terms of Standard Model couplings [Turok and Boyle, 2023]. Their amplitude is set by the high-temperature trace-anomaly coefficient c_β over an effective relativistic degree-of-freedom count; their tilt is set by the running of the dominant gauge coupling at the Planck scale.

This note reads that result through a signed-decomposition and cancellation-index vocabulary developed in a companion series on a different, purely mathematical partition construction [De Jesús, 2026e,b,d,a], used strictly as coordinates. The aim is organizational: to say precisely which structures the Turok–Boyle observables instantiate, and precisely where the analogy stops.

Non-claims. This note does not validate, extend, derive, or critique the Turok–Boyle construction. It does not claim that the signed-decomposition framework explains the primordial spectrum. It asserts no physical mechanism beyond what the cited authors state, and proposes no new cosmology. All physical inputs are theirs. It does not present κ_{cancel} as a new mathematical invariant: Section 5 identifies it exactly with a classical quantity from numerical linear algebra.

Relation to the first version. An earlier version of this note [De Jesús, 2026c] reported a cancellation index $\kappa_{\text{cancel}} \approx 0.20$ for the gauge sector and described it as characterizing that sector. That framing was too strong, and one sentence in it was simply wrong. The present version retains the descriptors but subordinates them to an invariance audit (Section 6), works out an explicit alternative decomposition (Section 7), and corrects the sign description (Caution 7.1). This third revision does not change any equation, theorem, or numerical value relative to the second; it corrects the sourcing of the physical inputs, adds the condition-number identification of κ_{cancel} , labels every algebraic construction as a witness rather than a physical

state, notes the contested status of the Turok–Boyle mechanism in the refereed literature, and updates the reproducibility appendix to the repository’s current, consolidated implementation.

2 The Turok–Boyle inputs, and their stated epistemic status

We quote only what is needed, and record how the authors themselves label each item and, where the provenance chain runs further back than their own paper, where it actually originates.

(TB1) The coefficient. The leading perturbative contribution to the high-temperature trace $T_\beta \equiv \langle T_\lambda^\lambda \rangle_\beta$ of the Standard Model is, in Turok and Boyle’s Eq. (4) [Turok and Boyle, 2023],

$$T_\beta^{\text{SM}} = \left(\frac{125}{108} \alpha_Y^2 - \frac{95}{72} \alpha_2^2 - \frac{49}{6} \alpha_3^2 \right) T^4 \equiv c_\beta T^4, \quad \alpha_i \equiv \frac{g_i^2}{4\pi}. \quad (1)$$

Turok and Boyle state that this is their Eq. (4), quoted from Arnold and Zhai [1995], Eq. (5.1) — the three-loop free energy of high-temperature QED and QCD with fermions. That is the coefficient’s true primary source, and Arnold and Zhai [1995] is cited directly here for the first time in this note’s history; the general SM group formula, applying Arnold and Zhai’s per-factor expression to each simple factor of the gauge group, is Turok and Boyle’s own step. The full Standard Model high-temperature pressure, including terms beyond gauge-plus-fermion, was computed separately by Gynther and Vepsäläinen [2006]. Under Turok and Boyle’s emergent-Higgs hypothesis no scalar channels appear in (1); Remark 3.2 records exactly what that omission costs relative to the textbook one-loop Standard Model beta functions.

(TB2) The couplings. At the Planck scale, $\alpha_Y = 0.0181$, $\alpha_2 = 0.0203$, $\alpha_3 = 0.0189$, giving $c_\beta = -0.0031$, with the $SU(3)$ contribution comprising 95% of the total [Turok and Boyle, 2023]. These couplings are not computed by Turok and Boyle themselves: they are taken from Buttazzo et al. [2013], via that paper’s own Ref. [37], and are cited to Buttazzo et al. [2013] directly for the first time in this note’s history.

(TB3) The amplitude. With $N_0 = 36$ dimension-zero fields,

$$\mathcal{P}_\mathcal{R} = \frac{3 \cdot 253}{7(2\pi)^4} \left(\frac{c_\beta}{N_{\text{eff}}} \right)^2, \quad N_{\text{eff}} = 106\frac{1}{4}. \quad (2)$$

This is a derivation internal to Turok and Boyle’s own framework, presented by them without hedging.

(TB4) The tilt. Focusing on the dominant $SU(3)$ contribution and leaving the flavour number free, Turok and Boyle observe that the coefficient factorizes as $T_\beta^{SU(3)} = -(12 + 5n_f)(11 - \frac{2}{3}n_f)\alpha_3^2 T^4/36$, with the second factor proportional to $\beta_{\alpha_3}/\alpha_3$ and the first accounted for by a thermal expectation $\langle F^2 \rangle_T$, and conclude

$$n_s - 1 = \frac{d \ln \mathcal{P}_\mathcal{R}}{d \ln k} = \frac{2\beta_{\alpha_3}}{\alpha_3} \approx -\frac{7\alpha_3}{\pi} = -0.042. \quad (3)$$

The authors explicitly label this a *heuristic analysis* and caution that it rests on key theoretical assumptions they have yet to verify: specifically, that the running factor depends on the spatial wavelength at which the dimension-zero field couples to the plasma, and a further final assumption about extrapolation from the Planck scale. Section 9 takes this at face value.

Contested status. The Turok–Boyle proposal is a February 2023 arXiv preprint with, as of this revision, no located journal version. Its central ingredient — dimension-zero scalar fields — has since been examined in the refereed literature by [Cline and Hell \[2026\]](#), who identify pathologies in the mechanism this note’s subject coefficient is meant to support. This note takes no position on that critique; the diagnostic below concerns what the two stated observables can resolve about a given decomposition, independently of whether the underlying mechanism survives scrutiny. A reader should know the proposal is contested, not merely unverified by its own authors.

3 The two-level structure of the coefficient

Turok and Boyle display the plasma/ β factorization only for the $SU(3)$ channel. It holds for all three, and stating it in that generality is what makes the rest of this note coherent.

Proposition 3.1 (Uniform positive-factor/ β -coefficient decomposition). *Write $b_a := \frac{11}{3}C_A^{(a)} - \frac{4}{3}S_F^{(a)}$ for the combination of group invariants using the data Turok and Boyle quote explicitly [[Turok and Boyle, 2023](#)] ($C_A = 0, 2, 3$ and $S_F = 5, 3, 3$ for $U(1)_Y$, $SU(2)_L$, $SU(3)_c$). Then each term of (1) factorizes exactly as*

$$c_a = -P_a \alpha_a^2 b_a, \quad P_{U(1)} = \frac{25}{144}, \quad P_{SU(2)} = \frac{19}{48}, \quad P_{SU(3)} = \frac{7}{6}, \quad (4)$$

with $b_{U(1)} = -\frac{20}{3}$, $b_{SU(2)} = \frac{10}{3}$, $b_{SU(3)} = 7$. All three factors P_a are strictly positive.

Proof. Direct substitution. For $SU(3)$: $b = \frac{11}{3} \cdot 3 - \frac{4}{3} \cdot 3 = 11 - 4 = 7$, and $-\frac{7}{6} \cdot 7 = -\frac{49}{6}$, reproducing the third coefficient of (1) and agreeing with $-(12 + 5n_f)(11 - \frac{2}{3}n_f)/36 = -49/6$ at $n_f = 6$. For $SU(2)$: $b = \frac{22}{3} - 4 = \frac{10}{3}$, and $-\frac{19}{48} \cdot \frac{10}{3} = -\frac{95}{72}$. For $U(1)_Y$: $C_A = 0$ gives $b = -\frac{20}{3}$, and $-\frac{25}{144} \cdot (-\frac{20}{3}) = +\frac{125}{108}$. $\square$

Remark 3.2 (b_a is the Arnold–Zhai gauge-plus-fermion combination, not the textbook one-loop Standard Model beta function). Each b_a above reproduces Turok and Boyle’s own (C_A, S_F) data exactly, and the values are not in question. They are *not*, however, the textbook Standard Model one-loop gauge beta coefficients, which in Standard-Model hypercharge normalization are $b_{U(1)} = -\frac{41}{6}$ and $b_{SU(2)} = \frac{19}{6}$. Each differs from the value used here by exactly $\frac{1}{6}$ — the contribution of a fundamental Higgs doublet, absent from the Arnold–Zhai gauge-plus-fermion high-temperature expression [[Arnold and Zhai, 1995](#)] because it is a QED/QCD-type computation with no scalar sector. This omission is not an inconsistency: it is exactly what Turok and Boyle’s own premise requires, since their construction needs the Higgs doublet to be *emergent* rather than fundamental. No equation or numerical value in this note is affected; the point is only that a reader who checks b_a against a textbook one-loop Standard Model beta function will find a mismatch, and should understand why.

Caution 3.3 (Algebraic factorization versus physical interpretation). For $SU(3)$, Turok and Boyle interpret the positive factor physically, through the thermal expectation $\langle F^2 \rangle_T$ of the gluon field strength, and it is that interpretation which motivates calling it a plasma factor. For $U(1)_Y$ and $SU(2)_L$, Proposition 3.1 establishes only the analogous algebraic factorization: P_a is defined as the ratio $-c_a/(\alpha_a^2 b_a)$, and its positivity is a computed fact, not a sourced physical identification. No claim is made here that $P_{U(1)}$ and $P_{SU(2)}$ are thermal expectation values, unless such an identification is supplied by the underlying thermal-field-theory calculation. We were unable to write the three values $25/144$, $19/48$, $7/6$ as a simple closed form in the group invariants (d_A, C_A, S_F); this is not a proof that none exists, but it is a reason not to present the algebraic rearrangement as a uniform physical structure. Nothing below depends on the interpretation — only on $P_a > 0$.

Corollary 3.4 (Where the signs come from). *Since $P_a > 0$ and $\alpha_a^2 > 0$, the sign of every channel is carried entirely by $-b_a$. The hypercharge channel is positive precisely because $C_A^{(U(1))} = 0$: an abelian factor has no gauge self-coupling, so its β -coefficient receives only the matter (screening) contribution. The two non-abelian channels are negative because antiscreening dominates. The mixed sign pattern of (1) is therefore not a numerical accident of the Planck-scale couplings; it is the abelian/non-abelian distinction.*

Proposition 3.1 exhibits c_β as a two-level object. Level 1 is the partition into three gauge factors. Level 2 resolves each factor into a positive prefactor times a β -coefficient which is itself a difference, $\frac{11}{3}C_A - \frac{4}{3}S_F$, of an antiscreening and a screening piece. Both levels are mathematically legitimate; Section 7 shows they give different answers for κ_{cancel} , and Section 6 explains why that had to happen.

4 Reading 1: what each observable actually reads

Observation 4.1 (Two different blindnesses). Relative to the structure of Proposition 3.1:

- (i) The amplitude (2) is an even function of the net, $\mathcal{P}_{\mathcal{R}} \propto c_{\text{net}}^2$, so it depends on $|c_\beta|$ and on nothing else about the decomposition. It reads all three channels, but only through the magnitude of their signed total; it is blind both to how that total is composed and to whether it is positive or negative. The latter blindness is not an artefact of bookkeeping: in [Turok and Boyle, 2023] the sign of c_β flips the sign of the coupling c_χ and hence of $\mathcal{R}$, but $\delta\chi$ is a zero-mean Gaussian field, so the sign is invisible in the two-point function that defines $\mathcal{P}_{\mathcal{R}}$.
- (ii) The tilt (3) is $2\beta_{\alpha_3}/\alpha_3 = -b_{SU(3)}/\pi$, so it reads the combination $b_{SU(3)}\alpha_3$, not the β -coefficient alone. It is independent of the positive prefactor $P_{SU(3)}$ and of the $U(1)_Y$ and $SU(2)_L$ channels entirely. This holds only under the heuristic wavelength-to-running assumptions of [Turok and Boyle, 2023].

Neither observable resolves the channel decomposition, and the two fail to resolve it in different directions: the amplitude aggregates across channels, retains both factors, and discards the sign; the tilt isolates one channel and discards its positive prefactor.

Caution 4.2 (The complementarity claim, corrected). The first version of this note described the amplitude and tilt as occupying complementary slots of one object, in the same diagnostic form as coupling/asymmetry pairs used elsewhere in the author’s own partition-construction series. That overstated the parallel in two ways, and the overstatement is now withdrawn. First, the pair is nested, not complementary: the tilt reads a strict sub-factor of what the amplitude reads, since $b_{SU(3)}$ appears in both, while the amplitude additionally carries P_a and the subdominant channels. Second, magnitude/logarithmic-derivative pairs are generic — every observable and its running form one — so recognizing that pattern here has limited discriminating power and should be read as shared vocabulary, not as evidence of shared structure.

5 The signed context and its descriptors

Definition 5.1 (Signed L^1 descriptors). For a finite family of signed contributions $\{c_i\}$ set

$$c_{\text{net}} = \sum_i c_i, \quad c_{\text{gross}} = \sum_i |c_i|, \quad \kappa_{\text{cancel}} = 1 - \frac{|c_{\text{net}}|}{c_{\text{gross}}} \in [0, 1], \quad (5)$$

with κ_{cancel} undefined when $c_{\text{gross}} = 0$. Note $\kappa_{\text{cancel}} = 1$ exactly when $c_{\text{net}} = 0$, i.e. under total cancellation; the physical case treated here has $c_{\text{net}} \neq 0$ and hence $\kappa_{\text{cancel}} < 1$. Writing

$P = \sum_{c_i > 0} c_i$ and $N = -\sum_{c_i < 0} c_i$ for the positive and negative masses,

$$\kappa_{\text{cancel}} = \frac{2 \min(P, N)}{P + N}. \quad (6)$$

Proof of (6). $c_{\text{net}} = P - N$ and $c_{\text{gross}} = P + N$, so $1 - |P - N|/(P + N) = (P + N - |P - N|)/(P + N) = 2 \min(P, N)/(P + N)$. $\square$

Identity (6) is the structural fact on which the whole of Section 6 turns: κ_{cancel} is a function of the two sign-class totals and of nothing else. It does not know how many channels there are, or which is which.

Proposition 5.2 (κ_{cancel} is a bounded reparameterization of the classical condition number of a sum). *Define $\text{cond} := c_{\text{gross}}/|c_{\text{net}}|$, for $c_{\text{net}} \neq 0$. Then*

$$\kappa_{\text{cancel}} = 1 - \frac{1}{\text{cond}}, \quad \text{cond} = \frac{1}{1 - \kappa_{\text{cancel}}}, \quad (7)$$

and both directions are exact. The quantity cond is precisely the factor in the standard error bound for floating-point summation, $|\widehat{\sum x_i} - \sum x_i| \lesssim (n-1)u \sum |x_i|$, restated relative to $|\sum x_i|$ [Higham, 1993]; it is universally called the condition number of the summation problem $\{c_i\} \mapsto \sum_i c_i$.

Proof. Immediate from Definition 5.1: $1 - 1/\text{cond} = 1 - |c_{\text{net}}|/c_{\text{gross}} = \kappa_{\text{cancel}}$, and solving for cond gives the second form. $\square$

Remark 5.3. Proposition 5.2 is the reason κ_{cancel} is not presented here as a new invariant. For $c_{\text{net}} \neq 0$, cond is always a finite real number in $[1, \infty)$ — the triangle inequality gives $c_{\text{gross}} \geq |c_{\text{net}}| > 0$, so $\text{cond} \geq 1$, with equality iff the family is sign-homogeneous — and $\kappa_{\text{cancel}} = 1 - 1/\text{cond}$ is an exact, invertible, monotone rescaling of that half-open interval onto $\kappa_{\text{cancel}} \in [0, 1)$; the value $\text{cond} = \infty$ is never attained by any actual finite family with $c_{\text{net}} \neq 0$, only approached in the limit of Theorem 6.2(vi). The boundary case $c_{\text{net}} = 0$, $c_{\text{gross}} > 0$ is separate, not a limit of this formula: there $\text{cond} = c_{\text{gross}}/|c_{\text{net}}|$ is undefined (division by zero), while $\kappa_{\text{cancel}} = 1$ exactly, directly from Definition 5.1 rather than from Proposition 5.2. The two descriptions agree in spirit — $\text{cond} \rightarrow \infty$ corresponds to $\kappa_{\text{cancel}} \rightarrow 1$ — but only the $c_{\text{net}} \neq 0$ case is literally a rescaling of a well-defined cond . The only thing gained by working with κ_{cancel} rather than cond directly, on their common domain, is boundedness. Every mathematical result below — Theorem 6.2 in particular — can be read equivalently as a statement about how the summation condition number of $\{c_i\}$ changes under refinement, coarsening, and relabelling of a fixed sum, which is the natural language in which numerical analysts already discuss such sums [Higham, 1993].

Proposition 5.4 (Marginal blindness, algebraic form). *Let $O = f(c_{\text{net}})$ for some f . Then:*

- (i) *O cannot distinguish two signed families with equal net, whatever their gross magnitudes or cancellation indices. In particular the amplitude assigns the same value to decomposition (1), with $\kappa_{\text{cancel}} = 0.20$, as to a single channel of the same net with $\kappa_{\text{cancel}} = 0$.*
- (ii) *If in addition f is even — as it is for the amplitude, $\mathcal{P}_{\mathcal{R}} \propto c_{\text{net}}^2$ — then O satisfies $O(c_{\text{net}}) = O(-c_{\text{net}})$ and is a function of $|c_{\text{net}}|$ alone, so it is blind to the overall sign as well as to the decomposition.*

Proof. (i) is immediate, since f is a function of c_{net} alone. (ii) is the definition of an even function. $\square$

Remark 5.5 (This proposition is nearly trivial, and should be labelled as such). Proposition 5.4 is an observation about functions of a sum: it is the elementary statement that a many-to-one observation map cannot be inverted, a standard non-identifiability fact [Rothenberg, 1971]. It carries no information beyond the definition of $\mathcal{P}_{\mathcal{R}}$. Its interest lies entirely in what it licenses one to ask — how much structure is hidden, and relative to what — and that question is answered by Sections 6 and 7, not by the proposition. It should not be cited as a result.

Remark 5.6 (The two blindnesses are logically independent). Sign blindness is not implied by decomposition blindness: an observable $O = c_{\text{net}}$ resolves the sign while remaining blind to the decomposition. Conversely, an observable of c_{gross} alone would be sign-blind while sensitive to the decomposition. The amplitude happens to suffer both. Note also that κ_{cancel} itself is invariant under the global flip $c_i \mapsto -c_i$, which interchanges P and N and leaves (6) unchanged; so the descriptors of Definition 5.1 restore the information the amplitude loses about the decomposition, but not the information it loses about the sign. The global sign flip used to establish this is an *algebraic witness*: it is not a physically available operation on the Standard Model coefficient c_β , whose sign is fixed once the theory and scale are fixed.

Remark 5.7 (Logical independence of amplitude and tilt is stronger than stated, but is not physical independence). The amplitude and tilt are also functionally independent on the three-coupling space, not merely independent as abstract functionals of different arguments. Treating $(\alpha_Y, \alpha_2, \alpha_3)$ as free: holding $c_\beta = -0.003082$ fixed and moving α_3 from 0.0189 to 0.0210, with α_Y solved to compensate (giving $\alpha_Y \approx 0.0303$), moves $n_s - 1$ from -0.042112 to -0.046792 ; holding α_3 fixed and moving α_2 over 0.018–0.023 moves c_β over -0.002966 to -0.003236 with the tilt exactly unchanged, since (3) does not depend on α_2 at all. So the two observables are genuinely functionally independent on the three-coupling space, not merely independent as abstract functionals. This is, however, a statement about the *observation map of the model family*: at the Standard Model Planck point the three couplings are determined, not free, so the witnesses above describe what the map $(\alpha_Y, \alpha_2, \alpha_3) \mapsto (c_\beta, n_s - 1)$ can and cannot separate, not a degeneracy among physically realized states. (The family is constrained: at $\alpha_3 = 0.017$ there is no real α_Y holding c_β fixed.)

Remark 5.8 (Signed L^1 is not a Fisher simplex). A companion construction on a different, purely mathematical partition [De Jesús, 2026b] proves a context-resolver theorem for non-negative context pieces $c_i \geq 0$ with $\sum_i c_i = c$, where a warped-product factor telescopes and subcomponent structure lives in Fisher information geometry on the resulting probability simplex. The positivity hypothesis is essential and fails here: by Corollary 3.4 the gauge decomposition is genuinely signed. Consequently the present statement is the signed analogue of marginal blindness, not an instance of it: it is algebraic (Proposition 5.4) rather than a telescoping identity, it comes with no Fisher-geometric content, and the descriptors of Definition 5.1 — the Jordan-decomposition total variation mass, not a Fisher metric — replace the simplex invariants. A Fisher treatment would require either normalizing the absolute contributions, which discards exactly the sign information at issue, or a genuine signed-context extension of the resolver theorem; that extension is not attempted here.

6 Invariance audit

This section asks, for each descriptor of Definition 5.1, what must be held fixed before the quantity is well defined.

Definition 6.1 (Refinement and coarsening). A family $\{c'_j\}$ refines $\{c_i\}$ if there is a surjection π from the index set of c' onto that of c with $c_i = \sum_{\pi(j)=i} c'_j$ for every i ; $\{c_i\}$ then coarsens $\{c'_j\}$. A refinement is *sign-homogeneous* if every group $\{c'_j : \pi(j) = i\}$ consists of terms of one

sign.¹

Theorem 6.2 (Monotonicity, and non-existence of a canonical value). *Let $\{c'_j\}$ refine $\{c_i\}$. Then*

- (i) c_{net} is unchanged;
- (ii) $c'_{\text{gross}} \geq c_{\text{gross}}$, with equality if and only if the refinement is sign-homogeneous;
- (iii) consequently $\kappa'_{\text{cancel}} \geq \kappa_{\text{cancel}}$, with equality under the same condition;
- (iv) κ_{cancel} is a symmetric function of the multiset $\{c_i\}$ and hence invariant under relabelling;
- (v) merging channels of opposite sign strictly decreases κ_{cancel} ;
- (vi) for any family with $c_{\text{net}} \neq 0$ and any $\varepsilon > 0$ there is a refinement with $\kappa_{\text{cancel}} > 1 - \varepsilon$, and coarsening to a single channel gives $\kappa_{\text{cancel}} = 0$. Hence over all decompositions of a fixed net, κ_{cancel} ranges over the whole of $[0, 1)$ and possesses no decomposition-free extremum.

Proof. (i) is the definition of refinement. (ii) By the triangle inequality $|c_i| = |\sum_{\pi(j)=i} c'_j| \leq \sum_{\pi(j)=i} |c'_j|$, with equality iff the terms in the group share a sign; summing over i gives the claim. (iii) $\kappa_{\text{cancel}} = 1 - |c_{\text{net}}|/c_{\text{gross}}$ is strictly increasing in c_{gross} at fixed $c_{\text{net}} \neq 0$, so (ii) transfers. (iv) is immediate from (6). (v) is (iii) read in reverse. For (vi), choose any component c_r and refine it as $c_r = (c_r + X) + (-X)$ for $X > \max\{0, -c_r\}$, which is a refinement in the sense of Definition 6.1 because the two pieces are assigned to c_r and sum to it. The net is unchanged, while the gross increases by $|c_r + X| + X - |c_r| = c_r + 2X - |c_r| \rightarrow \infty$ as $X \rightarrow \infty$, so $\kappa'_{\text{cancel}} \rightarrow 1$. Coarsening the whole family to $\{c_{\text{net}}\}$ gives $c_{\text{gross}} = |c_{\text{net}}|$ and $\kappa_{\text{cancel}} = 0$. More generally, for fixed $c_{\text{net}} = n \neq 0$ the two-component decompositions $(\frac{G+n}{2}, -\frac{G-n}{2})$, $G \geq |n|$, have net n and gross G , realizing every value $\kappa_{\text{cancel}} = 1 - |n|/G$ in $[0, 1)$. $\square$

Remark 6.3 (Every construction in Theorem 6.2(vi) is an algebraic witness, not a physical decomposition). The refinement $c_r = (c_r + X) + (-X)$ and the two-component family $(\frac{G+n}{2}, -\frac{G-n}{2})$ are constructed purely to demonstrate that κ_{cancel} has no decomposition-free supremum; they are not claimed to correspond to any physically realizable splitting of a Standard Model gauge-sector coefficient. The distinction matters because it is easy to conflate “ κ_{cancel} can be made arbitrarily close to 1 by *some* refinement” with “the gauge sector’s cancellation is arbitrarily close to total” — the theorem licenses only the first statement. The one refinement in this note that *is* physically motivated — gauge-boson versus matter, at fixed perturbative order and scale — is treated separately, in Section 7, precisely because it is not one of these algebraic witnesses.

Remark 6.4 (The threshold in (vi) is not cosmetic). The condition $X > \max\{0, -c_r\}$ is exactly what forces the refinement to cross the sign boundary. Refining the strong channel of (1), for which $c_r < 0$, with $X = 10^{-3} < -c_r$ leaves both pieces negative: the refinement is sign-homogeneous, the gross is unchanged, and κ_{cancel} stays at 0.1975, in accordance with (ii). Taking $X = 10^{-2}$ crosses the threshold and gives $\kappa_{\text{cancel}} = 0.829$; $X = 1$ gives $\kappa_{\text{cancel}} = 0.998$.

Theorem 6.2 is the reason the first version of this note overstated its case: it shows that κ_{cancel} is not a property of a physical system but of a stated resolution of that system, and that no refinement is distinguished by the formalism itself. Two further dependences, which are not decomposition choices at all, must also be fixed.

¹Precisely: a group is sign-homogeneous when no two of its nonzero terms have strictly opposite signs; adjoining zero terms changes nothing.

Scale	α_Y	α_2	α_3	c_{net}	κ_{cancel} (level 1)
Planck	0.0181	0.0203	0.0189	-3.08×10^{-3}	0.197
Electroweak	0.0102	0.0338	0.1184	-1.16×10^{-1}	0.002

Table 1: Scale dependence of the level-1 index, evaluating (1) with couplings at two scales. The electroweak row is illustrative of the running only: it applies the same high-temperature expression at a much lower scale, where the perturbative hierarchy and the applicability of the high- T expansion both differ, and the electroweak couplings used here are textbook values not drawn from a cited source (unlike the Planck row, which is Buttazzo et al. [2013] via Turok and Boyle [2023]). The point is the two orders of magnitude, not the second digit.

Truncation. Coefficients (1) are the leading $O(\alpha^2)$ perturbative contribution; the Standard Model free energy is known to higher order. At higher orders the expansion contains mixed-coupling terms ($\alpha_Y\alpha_2$, $\alpha_2\alpha_3$, and, absent the emergent-Higgs hypothesis, terms in the top Yukawa and the Higgs quartic), which belong to no single gauge factor. Beyond one loop the gauge beta functions of a general quantum field theory mix different gauge couplings, Yukawa couplings, and scalar couplings [Machacek and Vaughn, 1983]; the Standard Model’s own three-loop gauge beta functions have been computed explicitly and confirm this structure [Mihaila et al., 2012]. The one-channel-per-simple-factor decomposition therefore ceases to be closed and canonical beyond leading order: accommodating such terms requires either interaction channels indexed by pairs of factors, or a stated convention assigning each mixed term to one factor. This is not an impossibility — decompositions still exist — but it means the channel set is itself a further modelling choice, and by Theorem 6.2 that choice moves κ_{cancel} .

Renormalization scale. Each c_a depends on μ through $\alpha_a(\mu)$, and the three couplings run at different rates, so κ_{cancel} runs too. Table 1 records the effect.

Scheme and normalization. At one loop the b_a are scheme-independent, and the choice between Standard-Model and grand-unified normalization of hypercharge leaves c_a invariant, since $g_1 \mapsto \lambda g_1$ is compensated by $b_1 \mapsto b_1/\lambda^2$. Beyond one loop, scheme dependence enters through both the β coefficients and the extracted couplings. Basis changes are not a source of ambiguity here: the gauge group is a direct product of simple factors and each contributes a distinct gauge-invariant operator, so there is no mixing to choose. (This last stability would fail for extensions with several abelian factors, where kinetic mixing makes the abelian split basis-dependent.)

7 An alternative, physically meaningful decomposition

Theorem 6.2 says refinement can change κ_{cancel} . The relevant question is whether any *natural* refinement does. It does, and the refinement in question is displayed in the Turok–Boyle paper itself.

By Proposition 3.1, $c_a = -P_a\alpha_a^2(\frac{11}{3}C_A^{(a)} - \frac{4}{3}S_F^{(a)})$. The bracket is a difference of an antiscreening (gauge-boson loop) term and a screening (matter loop) term of opposite signs — exactly the split Turok and Boyle write out as $(11 - \frac{2}{3}n_f)$ for $SU(3)$. Resolving it gives the level-2 family

$$c_a^{\text{gauge}} = -P_a\alpha_a^2 \cdot \frac{11}{3}C_A^{(a)}, \quad c_a^{\text{matter}} = +P_a\alpha_a^2 \cdot \frac{4}{3}S_F^{(a)}. \quad (8)$$

This is a legitimate refinement at one loop, where b_a literally is that sum; it is diagrammatic bookkeeping, not a claim that the two pieces are separately observable.

Caution 7.1 (Correction of a factual error in the previous version). The first version of this note stated, in its abstract, in its main text, in a figure caption, and in its conclusion, that the

Operation	Effect on κ_{cancel}	Reason
Relabelling channels	unchanged	Theorem 6.2(iv)
Merging channels of like sign	unchanged	Theorem 6.2(ii)
Merging channels of opposite sign	decreases	Theorem 6.2(v)
Splitting into like-sign pieces	unchanged	Theorem 6.2(ii)
Splitting across the sign boundary	increases, un- bounded	Theorem 6.2(iii),(vi)
Global sign flip $c_i \mapsto -c_i$	unchanged	interchanges P, N in (6)
Change of basis	unchanged	direct-product gauge group; no mixing at this order
Hypercharge normalization (SM $\leftrightarrow$ GUT)	unchanged	$g_1 \mapsto \lambda g_1, b_1 \mapsto b_1/\lambda^2$
Renormalization scheme	unchanged at one loop	one-loop b_a scheme-independent
Loop order / truncation	changes, channel set not closed	Machacek and Vaughn [1983] , Mihaila et al. [2012]
Renormalization scale μ	changes, strongly	Table 1

Table 2: Invariance audit for the cancellation index. The same classification applies to c_{gross} ; c_{net} is unchanged by every operation in the first six rows and changes only with truncation and scale.

Level 1: gauge factors		Level 2: gauge-boson / matter	
Channel	c_a	Channel	value
$U(1)_Y$	$+3.792 \times 10^{-4}$	$U(1)_Y$ matter	$+3.792 \times 10^{-4}$
$SU(2)_L$	-5.437×10^{-4}	$SU(2)_L$ gauge	-1.196×10^{-3}
		$SU(2)_L$ matter	$+6.525 \times 10^{-4}$
$SU(3)_c$	-2.917×10^{-3}	$SU(3)_c$ gauge	-4.584×10^{-3}
		$SU(3)_c$ matter	$+1.667 \times 10^{-3}$
c_{net}	-3.082×10^{-3}	c_{net}	-3.082×10^{-3} (unchanged)
c_{gross}	3.840×10^{-3}	c_{gross}	8.479×10^{-3}
κ_{cancel}	0.197	κ_{cancel}	0.637

Table 3: The same coefficient at two resolutions, using the Planck-scale couplings of [Buttazzo et al. \[2013\]](#) as quoted in [Turok and Boyle \[2023\]](#). The net is identical by construction; the gross more than doubles and the cancellation index triples. The $U(1)_Y$ channel has no gauge-boson piece because $C_A = 0$ (Corollary 3.4).

hypercharge and weak contributions are sign-opposed and partially cancel before combining with the dominant strong contribution. This is incorrect. At level 1 the weak and strong channels carry the *same* sign; only hypercharge is positive. Moreover a sum has no sequencing, so “before combining” has no content, and by (6) the index is $\kappa_{\text{cancel}} = 2c_{U(1)}/c_{\text{gross}}$: it is determined entirely by the single positive channel and says nothing whatever about any cancellation between the weak and strong channels. The correct level-1 statement is: one channel of three is positive, it carries 9.9% of the gross magnitude, and κ_{cancel} is twice that fraction.

Remark 7.2 (Why the refinement does not empty the diagnostic). Three things survive Theorem 6.2 and Table 3 (Figure 1). First, the direction of the effect is fixed, not arbitrary: by Theorem 6.2(iii) refinement can only raise κ_{cancel} , so level-1 values are lower bounds on the cancellation visible at any finer resolution of the same object, and the statement “at least a fifth of the mobilized gauge magnitude does not appear in the net, at the coarsest gauge-factor resolution” is well posed and true. Second, the blindness statement strengthens rather than weakens under refinement: the amplitude is a function of c_{net} , which Theorem 6.2(i) shows is

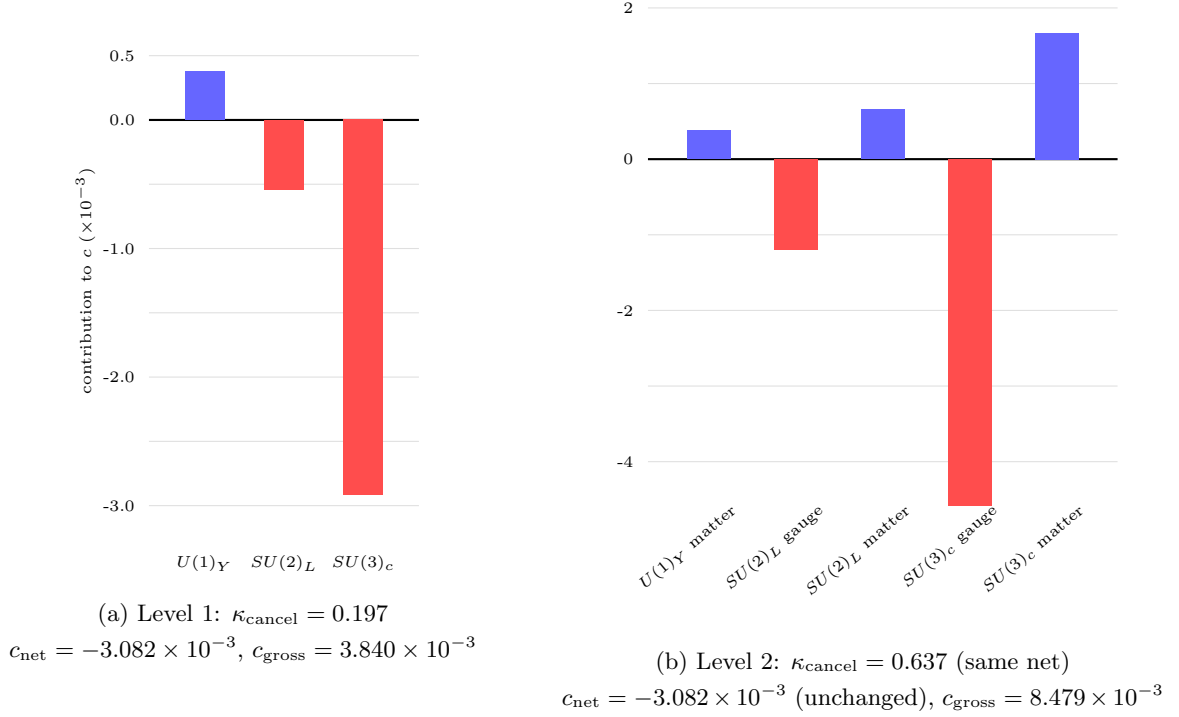


Figure 1: The trace-anomaly coefficient at two resolutions. (a) The three gauge factors: $\kappa_{\text{cancel}} = 0.197$, and the entire index is carried by the single positive $U(1)_Y$ channel, since $\kappa_{\text{cancel}} = 2 \min(P, N)/c_{\text{gross}}$ and $P = c_{U(1)}$. (b) The gauge-boson/matter refinement of (8): identical net, larger gross, $\kappa_{\text{cancel}} = 0.637$. Neither value is more correct than the other; each describes a stated resolution. Blue denotes positive contributions, red negative. Values as in Table 3.

invariant under the entire refinement hierarchy, so the amplitude is blind not to one alternative decomposition but to all of them simultaneously. Third, the pair $(c_{\text{net}}, \kappa_{\text{cancel}})$ remains a legitimate descriptor of a *stated* resolution, in the way that a coarse-graining scale is a legitimate parameter rather than an embarrassment. What must be abandoned is the reading of κ_{cancel} as a number attaching to the Standard Model; what replaces it is κ_{cancel} as a resolution-depth coordinate: the pair $(0.20, 0.64)$ for levels 1 and 2 is more informative than either number alone, because the gap between them is the amount of sign structure that the gauge-factor labelling conceals — unlike the algebraic witnesses of Remark 6.3, this gap is realized by a refinement of the same physical expression at fixed order and scale, and is therefore the one genuinely physical comparison available at this order.

8 Consistency of the two readings

The first version of this note contained a tension it did not acknowledge. Reading 1 was built on the plasma/ β factorization; Reading 2 computed κ_{cancel} on the unfactorized gauge channels. But the β factor is exactly where the additional sign structure lives (Section 7), so the first reading endorses a resolution the second ignores. One cannot celebrate the factorization and then treat the unfactorized channel as the natural unit without argument.

The present version resolves this by placing both readings on the two-level structure of Proposition 3.1 from the outset. Level 1 supports the amplitude’s aggregation statement; level 2 supports the tilt’s factor-selection statement and supplies the refinement of Section 7. Observation 4.1 is then a single statement about one object at a stated resolution, and Table 3 reports both resolutions rather than silently choosing one. The unified reading is:

Both Turok–Boyle observables are projections of a two-level signed structure that

neither of them resolves. The amplitude projects onto the magnitude of the signed sum across level-1 channels; the tilt projects onto one level-2 factor of one level-1 channel. The descriptors c_{gross} and κ_{cancel} quantify what is lost in the first projection, relative to a stated resolution, and are monotone in that resolution.

9 Epistemic tiers

Tier 1 — exact algebra, given the stated inputs. Equation (1) as an expression, and the numerical values of Tables 3 and 1 computed from it. The uniform algebraic factorization $c_a = -P_a \alpha_a^2 b_a$ with $P_a > 0$ (Proposition 3.1) and the sign attribution of Corollary 3.4. (The thermal reading of P_a is Tier 2 for $SU(3)$ and is not claimed at all for the other two channels; see Caution 3.3.) The descriptor identities (5), (6). The condition-number identity of Proposition 5.2. The monotonicity theorem (Theorem 6.2) and the invariance audit of Table 2. The algebraic marginal-blindness of Proposition 5.4, with the caveat of Remark 5.5. That the signed decomposition is not a Fisher simplex (Remark 5.8).

Tier 2 — results imported from Turok and Boyle under their own stated assumptions. That $\mathcal{P}_{\mathcal{R}} \propto (c_{\beta}/N_{\text{eff}})^2$, a derivation internal to a framework this note does not assess. And, in particular, the tilt relation (3) and the plasma/ β interpretation of the two factors, which Turok and Boyle explicitly present as heuristic and dependent on assumptions they state they have not verified. Any diagnostic reading of the tilt inherits that status.

Tier 3 — structural interpretation offered here, not established. The reading of the amplitude and tilt as two blindnesses of one object (Observation 4.1); the reading of κ_{cancel} as a resolution-depth coordinate (Remark 7.2); the suggestion that the level-1/level-2 gap is a meaningful quantity rather than an artefact of a convenient split. These are organizational proposals, falsifiable only in the weak sense that a better-motivated canonical decomposition, if one existed, would displace them.

Tier 4 — not claimed, and explicitly disavowed. Any physical equivalence between the signed-decomposition framework and the Turok–Boyle construction. Any derivation, validation, extension, or critique of their result — including any position on the pathologies identified by Cline and Hell [2026]. Any Fisher-geometric treatment of the signed decomposition absent an explicit signed-context extension of the resolver theorem. Any reading of κ_{cancel} as a property of the Standard Model, of the gauge sector, or of any physical system rather than of a stated decomposition at a stated scale and truncation. Any reading of κ_{cancel} as a newly discovered mathematical invariant, rather than the reparameterization of Proposition 5.2. Any suggestion that the numerical closeness of κ_{cancel} to $1/5$ is meaningful.

10 Limitations

(L1) The formalism selects no unique resolution. Definition 5.1 does not privilege any decomposition, and Theorem 6.2(vi) shows κ_{cancel} spans $[0, 1)$ over decompositions of a fixed net; hence no decomposition-independent value of κ_{cancel} is established here. This is a statement about the formalism, not about physics: the gauge-factor split is unusually natural, since the gauge group is a direct product and each factor contributes a separately gauge-invariant operator, and it remains entirely possible that a physical argument privileges one resolution. We claim only that the present framework supplies no such argument, and that Section 7 exhibits a competing split drawn from the source paper itself.

- (L2) **The descriptors are truncation-bound.** At higher order the decomposition into independent single-gauge-factor channels ceases to be closed and canonical (Machacek and Vaughn, 1983, Mihaila et al., 2012), because mixed-coupling terms require either additional interaction channels indexed by pairs of factors or a convention for assigning them to one factor. Either route is itself a resolution choice, and by Theorem 6.2 introducing interaction channels of either sign can only leave κ_{cancel} unchanged or raise it.
- (L3) **The descriptors are scale-bound.** Two orders of magnitude separate the Planck and electroweak values (Table 1).
- (L4) **The channel set is model-bound.** Under the emergent-Higgs hypothesis of Turok and Boyle [2023], no scalar channels appear at this order; see Remark 3.2 for exactly what that costs relative to the textbook Standard Model beta functions. That is an assumption of their proposal, not a fact about the Standard Model, and relaxing it adds channels and changes c_{gross} and κ_{cancel} .
- (L5) **The marginal-blindness proposition is near-trivial.** See Remark 5.5. The framework’s contribution is vocabulary and bookkeeping, applied to a specific physical coefficient, not a theorem.
- (L6) **The complementarity analogy is weaker than in the companion construction.** See Caution 4.2: the pair is nested rather than complementary, and magnitude/log-derivative pairs are generic.
- (L7) **Non-identifiability here is a statement about this diagnostic, not about physical consistency.** That the amplitude and tilt cannot jointly recover the signed channel decomposition, gross magnitude, or sign of c_β does not indicate any inconsistency in the Turok–Boyle construction; it indicates that those two particular observables, as defined, are many-to-one functions of the underlying sum. Whether some other observable resolves more of the decomposition is a separate, open question this note does not address.
- (L8) **Algebraic witnesses are not alternative physical realizations.** The two-component construction of Theorem 6.2(vi), the global sign flip of Remark 5.6, and the coupling-space witnesses of Remark 5.7 are constructed to prove mathematical statements about the descriptors and the observation map. None of them describes an alternative Standard Model state, and none should be read as such. Section 7’s gauge-boson/matter refinement is the one physically motivated comparison in this note.
- (L9) **The underlying mechanism is contested in the refereed literature.** Cline and Hell [2026] identify pathologies in the dimension-zero scalar field construction that the Turok–Boyle trace-anomaly analysis depends on. This note takes no position on that critique and does not need to: the diagnostic below concerns what two stated observables reveal about a decomposition, independently of whether the mechanism producing that decomposition survives further scrutiny.
- (L10) **This note assesses no physics.** It records a coordinate reading of an external result whose own status is contested and, in the tilt sector, explicitly provisional by its authors’ own account.

11 Conclusion

The trace-anomaly coefficient of Turok and Boyle [2023] is a signed, two-level object, ultimately sourced from Arnold and Zhai [1995] for its functional form and Buttazzo et al. [2013] for its numerical couplings: three gauge-factor channels, each factorizing exactly into a positive

prefactor — thermal in origin for $SU(3)$ on the authors’ reading, algebraically defined for the other two (Caution 3.3) — and a one-loop-type β -coefficient combination that is itself a difference of antiscreening and screening parts, and that coincides with the textbook one-loop Standard Model beta function only because no fundamental Higgs doublet contributes (Remark 3.2). The amplitude reads the magnitude of the signed sum over level-1 channels, being an even function of it, and so discards both the decomposition and the overall sign. Under the heuristic assumptions of Turok and Boyle [2023], the tilt reads the combination $b_3\alpha_3$ from one level-2 factor of one level-1 channel, independent of that channel’s positive prefactor and of the subdominant channels. Neither resolves the decomposition, and their blindnesses are of different kinds.

The descriptors that quantify what is unresolved are classical, not new: c_{net} and c_{gross} are the Jordan-decomposition total signed and total variation masses of a discrete signed measure [Bogachev, 2007], and κ_{cancel} is an exact, bounded reparameterization of the condition number of a sum [Higham, 1993] (Proposition 5.2). They are honest bookkeeping but not invariants. Only the net is fixed once scale and truncation are fixed; the gross magnitude and the cancellation index additionally require a stated decomposition, are monotone non-decreasing under refinement, and range over all of $[0, 1)$. Concretely, the value $\kappa_{\text{cancel}} \approx 0.20$ at the gauge-factor resolution becomes 0.64 under the gauge-boson/matter refinement that Turok and Boyle [2023] itself displays, and 0.002 at the electroweak scale. The first version of this note reported 0.20 as though it characterized the Standard Model gauge sector; it does not, and that claim is withdrawn.

What remains is worth stating precisely, and is all that is claimed. A net-only observable is invariant under the entire refinement hierarchy of a signed decomposition, and therefore blind to arbitrarily deep sign structure, provided that structure is reached by algebraic witnesses rather than physically motivated refinements; κ_{cancel} is a monotone coordinate on that hierarchy, in exact correspondence with the classical summation condition number, rather than a property of the system; and the gap between the two physically meaningful resolutions available here — gauge factors and gauge-boson/matter, 0.20 to 0.64 — measures how much sign structure a chosen labelling conceals. The contribution is the systematic application of this classical vocabulary, and the resulting identifiability audit, to one specific cosmological coefficient whose combination of decomposition-, scale-, and truncation-dependence had not previously been examined this way. It is not new mathematics, it is not new physics, and it is not a result about the Standard Model or about the validity of the Turok–Boyle construction, which remains contested (Cline and Hell, 2026; Limitation L9).

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A Reproducibility

All numbers in this note follow from (1) and the couplings of Table 1. A reference implementation accompanies this manuscript at github.com/innerlightr-wq/signed-context-decomposition-cosmology. The canonical module is the single top-level file `signedctx.py`; an earlier layout also carried a second, superseded copy under `src/`, which the test harness imported by mis-

take under `pytest` while the canonical module ran interactively — this has been corrected, the duplicate removed, and a regression test now guards against it recurring (see the repository’s `docs/CODE.ARCHITECTURE.md`). No mathematics, physics input, or published number changed as a result of that correction.

```
git clone \
    https://github.com/innerlightr-wq/signed-context-decomposition-cosmology.git
cd signed-context-decomposition-cosmology
pip install -r requirements.txt    # numpy, pytest

python signedctx.py                # self-check against this note’s tables
python signedctx.py --levels       # both resolutions side by side
pytest -q                          # 62 tests, canonical module only
```

What is exact algebra versus what is numerical. The identities (5)–(7), the monotonicity theorem (Theorem 6.2), and the sign attribution of Corollary 3.4 are proved in this note and additionally checked symbolically/numerically by the module’s self-check. The entries of Tables 3 and 1 are floating-point evaluations of exact rational coefficients at the quoted couplings and are exact to the precision shown; the module reports them to full double precision. Example output:

```
level 1: net=-3.081767e-03  gross=3.840143e-03  kappa=0.197482
level 2: net=-3.081767e-03  gross=8.478969e-03  kappa=0.636543
```

The $SU(3)$ share of $|c_{\text{net}}|$ is 94.7%, consistent with the 95% quoted by [Turok and Boyle \[2023\]](#).

Theorem 6.2(vi), a second witness. Remark 6.4 illustrates unbounded growth of κ_{cancel} by splitting the strong channel itself. An independent witness, using the same three level-1 values c_1, c_2, c_3 of Table 3 with an *injected*, unrelated canceling pair $(+X, -X)$ adjoined to the family (a refinement of the zero vector, not of any existing channel), gives the same qualitative growth:

```
X=1e-3: kappa([c1,c2,c3,+X,-X]) = 0.4723
X=1e-2: kappa([c1,c2,c3,+X,-X]) = 0.8707
X=1e-1: kappa([c1,c2,c3,+X,-X]) = 0.9849
```

independent of how the pair is attributed to a channel, confirming Theorem 6.2(vi) by a second, structurally different construction. As with every construction in that theorem (Remark 6.3), this is an algebraic witness, not a physical decomposition.

What is exploratory and not part of any claim in this note. The repository’s third example script runs an illustrative ensemble diagnostic over four *hypothetical* gauge-sector completions, to show what an ensemble test of Limitation (L1) would look like. It certifies nothing — a single physical system cannot populate such an ensemble — and no number from it appears in this manuscript.

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